

Energy consumption estimates

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I. INTRODUCTION

This will be filled in by Jinzhao.

II. SIMULATION RESULTS

Alongside mathematical simulations, we chose to leverage library simulations of both NISQ era circuits, using full density matrix simulations with standard Qiskit noise models, and FTQC era circuits where we employ clifford gates and Pauli channel noise models to allow for stabiliser state tracking.

A. NISQ results

Let me go through a brief overview of how results are generated:

- The necessary Qiskit QuantumCircuit objects are generated for each algorithm being run. No noise is yet added. These circuits include many trials and many shots of each trial.
- The circuits are transpiled to the relevant basis gates using standard qiskit transpilation and tunable optimisation (default level=3)
- The circuits are ran on a noisy AerSimulator object. Sometimes it is also run on a noiseless AerSimulator for comparison.
- A performance value is obtained from averaging over all shots and trials

The validity of these results hinges primarily upon the correctness of the code used to generate them. Thus, we have generated an extensive suite of unit tests to ensure the framework is operating as expected.

Moreover, the validity and applicability of these results further hinges upon the use of correct and properly catalogued assumptions used to generate them. We employ the following to mitigate the risks of compromising these qualities:

- The use of a variety of different methods of producing results, including a variety of different algorithms and methods of generating noise models. These methods vary in complexity and specificity, avoiding how many assumptions are universally relied upon.
- Comprehensively cataloguing the assumptions used at every stage of generating the results, including how they change for each method of producing results.

To do this, we treat each set of results as a case study. Because there is so much overlap between the assumptions of each case study, we include a set of all set of assumptions in the appendix, and specify which ones apply for each.

Something important to emphasise is the flexibility and customisability of this framework. To fully understand the power

Let us now go through the general assumptions that apply to all the case studies below:

- All noise models are inherently gate-local (and thus qubit-local)
- All noise models are state independent and unchanging over time
- All noise model are inherently Markovian

1. Case Study 1: *Quantum Volume, Simple Arrhenius Noise Model*

The simplest case study, where we use a simple Arrhenius noise model on a simulated quantum volume metric. Assumption set:

2. Hardware Parameters

I make the following modelling assumptions for the IBM processors:

We use the following hardware parameters in the temperature dependent model:

- γ_0 is fit at runtime to the known values of T_1 at the default operational mixing chamber temperature
 - Assume the operational cryostat temperature to be 0.013K
- $\gamma_{\varphi-\text{base}}$ is fit at runtime to the known values of T_2 at the default operational mixing chamber temperature
 - Again, assume the operational cryostat temperature to be 0.013K
- $\frac{\gamma_{MXC}}{\gamma_0}$ is 0.85.
 - This represents the maximum value from the experimental dataset reported in Ref. [1].
- T_{env} as 30mK
 - Effective temperatures as low as 20mK have been observed Ref. [2].

- Δ (superconducting energy gap) of $2.88e^{-23}$
 - This is the known value for Aluminium, which we can assume the electrodes used in IBM transmon qubits are made of
- ω_{ge} is extracted from known configuration data for each backend individually for each qubit
- N_e , the number of effective channels in the Josephson Junction, is estimated at 1.29×10^5 . We calculate this value using:
 - An estimated normal resistance of the Josephson Junction of $5k\Omega$
 - Transparency of the junction in the subgap regime of 5×10^{-5}

- E_c , charging energy of the capacitors in the transmon qubits, of $0.95\mu\text{eV}$
- E_j , josephson energy of the transmon qubits, of $62.1\mu\text{eV}$

Starting with an estimated value of γ_0 : $5e3$, and an estimated standard operational cryostat temperature of 0.013K , we minimise the squared error between the calculated and known, default relaxation time of the qubits. We then do the same for the value of $\gamma_{\varphi\text{-base}}$ keeping γ_0 constant

The final scale factors for the value of T_1 and T_2 for each qubit are stored and applied to the calculated values to maintain granularity

When calculating gate errors, the difference between the known gate error and calculated gate error at the operational temperature is stored and assumed to be representative of the non-thermal error, and thus independent of temperature. It is then added to future calculated gate errors

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- [1] D. S. Lvov, S. A. Lemziakov, E. Ankerhold, J. T. Pelttonen, and J. P. Pekola, Physical Review Applied **23**, 054079 (2025).
- [2] S. Simbierowicz, M. Borrelli, V. Monarkha, V. Nuutinen,

and R. E. Lake, Physical Review X Quantum **5**, 030302 (2024).